Zhiyao Lei

Amsterdam, Netherlands — leizhiyao2005@gmail.com — +31 6 1644 8368

Education

Sep 2023 – present	**University of Amsterdam**, Amsterdam, Netherlands BSc Econometrics and Data Science Expected graduation: July 2026; Current GPA: 8.2/10
Aug 2023 – Sep 2023	Preparation for undergraduate studies and relocation to the Netherlands
Sep 2020 – Jul 2023	**Wuhan Britain China School**, Wuhan, China GCE Advanced Level, Cambridge International A-levels: Mathematics A*, Further Mathematics A*, Physics A*, Economics A

Relevant Coursework

- **Mathematics and Statistics:** Advanced Linear Algebra, Multivariable Analysis and Optimization, Mathematical Statistics, Numerical Analysis, Dynamical Systems
- **Econometrics and Data Science:** Econometrics, Time Series Analysis, Statistical Learning, Algorithms and Data Structures
- **Machine Learning and AI:** Reinforcement Learning, Text Retrieval and Mining / Natural Language Processing

Selected Projects and Coursework

- **Bachelor's Thesis: Fair Representation Learning and Linear Concept Erasure**
  Studying fairness–utility trade-offs through continuous relaxation of linear concept erasure methods, using constrained optimization, covariance-based fairness constraints, and synthetic experiments.
- **Reinforcement Learning**
  Implemented and analysed tabular RL algorithms, including bandits, dynamic programming, Monte Carlo methods, temporal-difference learning, and planning methods on finite MDPs.
- **Time Series Forecasting**
  Applied ARIMA/Box–Jenkins modelling, forecast comparison, out-of-sample evaluation, and volatility modelling with ARCH/GARCH.
- **Machine Learning and NLP**
  Worked on regularized regression, splines/GAMs, tree-based ensembles, text preprocessing, topic modelling, embedding-based methods, BERT, and Transformer architectures.
- **Kaggle Crypto Market Prediction**
  Built baseline models using XGBoost and LightGBM, with feature engineering, feature selection, and correlation-based validation.

Academic Activities

Aug 2025	**Summer School in Machine Learning and Optimization**, University of Jyväskylä, Finland Coursework in stochastic control and interactive multi-objective optimization, including Bellman equations, MDPs, PDE/BSDE methods with deep learning, scalarization, and Pareto exploration.

Technical Skills

Programming	Python, R, LaTeX, Git
Libraries	NumPy, SciPy, pandas, scikit-learn, matplotlib, PyTorch, HuggingFace Transformers
Methods	Regression, regularization, cross-validation, gradient boosting, time series analysis, representation learning, reinforcement learning
Mathematics	Linear algebra, optimization, probability and inference, Markov decision processes

Languages

Chinese	Native
English	Fluent